

Sports Field Booking Information System Based on Web: Analysis and Design Approach

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ABSTRACT

The process of booking sports fields that still relies on manual methods often causes various obstacles, such as administrative inefficiencies, a high potential for recording and scheduling errors, and limited access to information for users. The gap between the expectation of a fast, accurate, and transparent ordering system and the reality of existing manual systems drives the need for innovative solutions. This research aims to design a web-based sports field booking information system through a systematic system analysis and design approach, as a direct response to these problems. The methodology used is a comprehensive literature study, including the identification of user and manager needs, the collection of scientific references, in-depth analysis of previous research, and the synthesis of findings as the basis for design. This approach ensures that the resulting design is not only innovative but also builds on existing best practices, thereby increasing the relevance and potential for successful implementation. The result of this research is a comprehensive information system design model, including essential features such as user registration and login, search and field selection, a real-time booking process integrated with online payment systems, automatic confirmation, booking history, and data management modules for managers. The design of the system is also supported by detailed design artifacts such as the Use Case Diagram, Sequence Diagram, Deployment Diagram, Entity-Relationship Diagram (ERD), and an intuitive user interface (UI) design.

Keywords: information systems; sports field booking; analysis and design; web; administrative efficiency

INTRODUCTION

The public's need for sports facilities is increasing along with the development of healthy lifestyles and growing awareness of the importance of physical activity. However, the process of booking sports fields, which is still done manually, often causes various problems, such as inaccurate record keeping, scheduling errors, and limited information accessibility for users. This condition leads to low efficiency and effectiveness in administrative management, potential schedule conflicts, and decreased service quality for users (Rahmi & Pramono, 2019). Ideally, the sports field booking process should be conducted quickly, accurately, and transparently, allowing users to easily obtain information on court availability, make reservations, and receive real-time confirmations. However, in reality, the manual systems still implemented in many places have not been able to meet these needs. This indicates a gap between ideal conditions and the actual situation in practice (Zhao et al., 2017).

Various previous studies have shown that the application of information systems can increase efficiency and accuracy in data management and service delivery (Munif et al., 2019). In addition, the use of digital-based information systems is also able to minimize the occurrence

of human error and accelerate administrative processes (Maulana et al., 2015). By adopting a system analysis and design approach, the information system design process can be carried out systematically to produce solutions that meet the needs of both users and sports field managers (Anwar et al., 2020).

While the adoption of digital technologies has been shown to increase operational efficiency and improve service quality in various sectors, the sports facility booking industry has been relatively slow to implement modern automated systems (Zhao et al., 2017). Previous studies, such as those by (Munif et al., 2019) and (Maulana et al., 2015), have demonstrated the effectiveness of information systems in enhancing service delivery and reducing administrative overhead. Despite these advancements, there remains a notable gap in research specifically focused on the design and implementation of web-based sports field booking information systems that integrate user-friendly interfaces with real-time data management and online payment solutions.

The urgency of addressing this gap is heightened by the need for improved efficiency, transparency, and customer experience in managing sports facility bookings. This research aims to fill this gap by designing a web-based sports field booking information system that ensures ease of use, real-time booking capability, and online payment integration. The novelty of this study lies in its systematic design approach, which builds upon existing best practices while integrating innovative features tailored to the specific needs of both users and facility managers.

By adopting a systematic approach to system design, this study contributes to the development of an efficient and user-friendly platform that minimizes booking errors and administrative burdens. The findings from this research can provide valuable insights for sports facility managers seeking to implement digital solutions, thereby enhancing customer satisfaction and streamlining booking processes. Ultimately, this research contributes to the broader field of digital transformation in facility management and may serve as a model for similar systems in other sectors.

Based on this background, this study aims to design a sports field booking information system using a system analysis and design approach. The results of this research are expected to provide innovative solutions in the management of sports field bookings, improve service quality, and provide convenience for users to make reservations online.

RESEARCH METHOD

This study used a literature review method as the main approach in designing a sports field booking information system. The literature review was conducted by collecting, analyzing, and summarizing scientific articles, journals, and other reliable sources related to booking information systems, sports field management, and software design methodologies. This approach aimed to obtain a comprehensive understanding of user needs, relevant system features, and best practices in the development of booking information systems.

The research process began with identifying the needs of users and sports field managers to determine the main features required in the system. Subsequently, references were collected through a review of journals, articles, and conference proceedings related to booking information systems, sports field booking systems, and software development methods. Several key references included studies on the design and implementation of web-based court booking systems (Safi et al., 2019), web-based futsal court booking information systems (Iriadi et al.,

2019), Android-based booking systems (Ratnasari et al., 2018), and badminton court rental information systems (Resmana, 2017).

The next stage involved analyzing previous studies to identify commonly implemented features, system development approaches, and challenges encountered in similar systems. The findings from this literature review were then synthesized to form the basis for the proposed system design.

At the system design stage, the overall structure of the system was developed, including business process flows, identification of system actors, and the design of the user interface. The system architecture was designed using a web-based client–server approach, with centralized data storage on the server to support system reliability and maintenance. System functionality and actor interactions were represented using use case diagrams and sequence diagrams. Data structure and relationships were modeled using an Entity Relationship Diagram (ERD), while the user interface design emphasized usability and responsiveness (Putra et al., 2018).

All design components were then integrated into a comprehensive system model to ensure that the booking process, data management, and user interaction could operate effectively (Nugraha et al., 2017). The design was evaluated by comparing it with approaches and system features reported in previous studies to assess its feasibility and relevance. The results of this research contribute to the design of a sports field booking information system intended to improve management efficiency, reduce administrative errors, and facilitate online reservations for users. The proposed design may also serve as a reference for the future development and implementation of similar systems in sports facilities.

RESULT AND DISCUSSION

Based on the results of literature studies and analysis of various studies related to the booking information system and the sports field booking information system, a web-based sports field booking information system model has been designed which has been developed by paying attention to the needs of users and managers, and adopting the main features of previous research. Key features implemented in this system include user registration and login to enable access to booking services, search and field selection based on date and time according to the user's wishes, real-time ordering process supported by online payments to improve transaction efficiency, automatic confirmation along with order history so that users can monitor orders that have been placed, as well as field data management and schedules that make it easier managers in updating information regularly. (Ratnasari et al., 2018) (Ratnasari et al., 2018)

Use Case Diagram

In a sports pitch booking system, a use case diagram is used to illustrate the interaction between actors and the system as well as the main functions available. The actors involved in this system include guests, members, employees, admins, and managers, each of whom has different access rights. Guests can do field searches and registrations, while members can login, place booking, pay, view booking history, and provide ratings. Employees have access to login, manage payments, schedules, and bookings, while admins have full access rights such as managing users, payments, schedules, fields, reports, and bookings. Managers can also sign in, manage payments, schedules, reports, and bookings. With this use case diagram, the needs of each actor and the flow of business processes in the system can be clearly identified, so that the

development of the system becomes more directed and efficient, as also explained in the research on the importance of mapping user needs and the separation of access rights in the sports field rental system. Stuttgart & Abdillah, 2018)

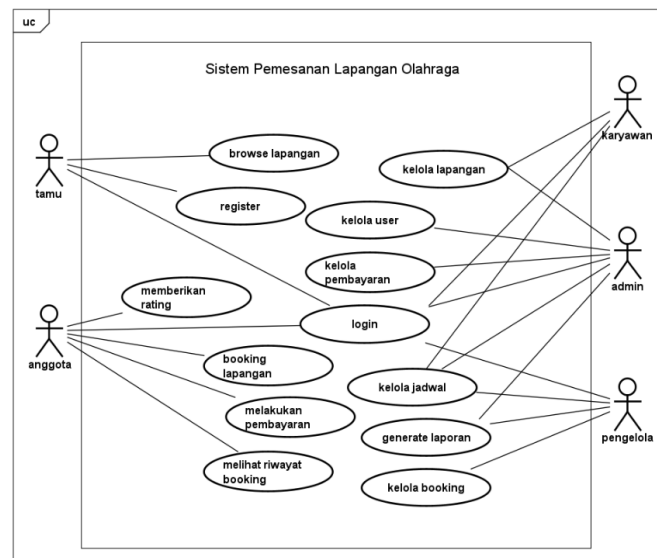


Figure 1. Use case diagram

Sequence Diagram

This diagram illustrates the process of booking a field by a Member through several stages, starting from opening the field list page, where the system retrieves and displays field data from the database. Once members have selected one of the fields, the system displays the details of the fields along with the available schedules. Next, members select a schedule and fill in the booking data, then send the booking request to the system. The system will process and store the booking data to the database; If the schedule is still available and the data is valid, the system confirms the booking to the member, but if the schedule is full or an error occurs, the system displays an error message. This diagram emphasizes the flow of interaction between users, application interfaces, backend systems, and databases to ensure the ordering process runs automatically and avoids double bookings.

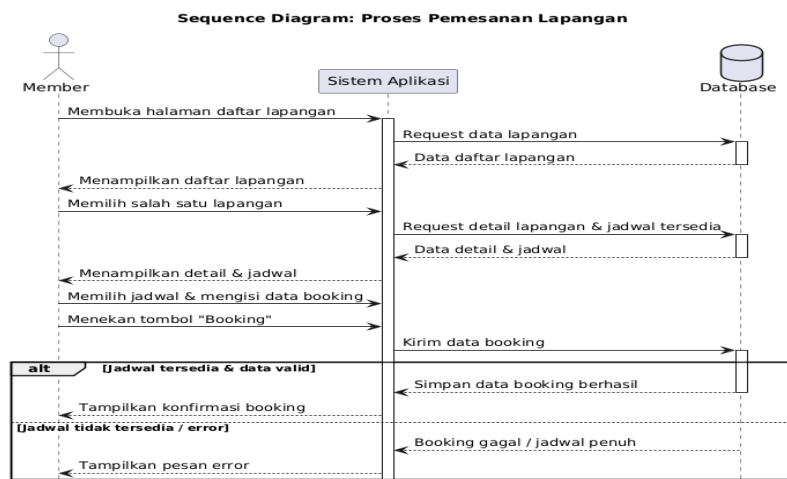


Figure 2. Sequence diagram of the field booking process by members

This chart shows the payment flow made by a member after placing an order. The process begins when the Member opens the payment page (PaymentPage), which then requests the payment details and payment method to the System. The system retrieves the booking data along with the payment method options from the Database (DB), and displays it on the payment page for the Member to select. Once the Member selects a payment method and confirms, the payment data is sent to the System, which then processes the transaction through the Payment Gateway. The Payment Gateway returns a notification of the payment status (successful or failed) to the System. If the payment is successful, the System updates the booking status to "Paid" in the Database, then displays a successful notification to the Member. If the payment fails, the System displays a failed notification and asks the Member to repeat the payment. This diagram emphasizes integration between applications, payment gateways, and databases to ensure that the payment process runs automatically, securely, and the booking status is always updated according to the transaction results.

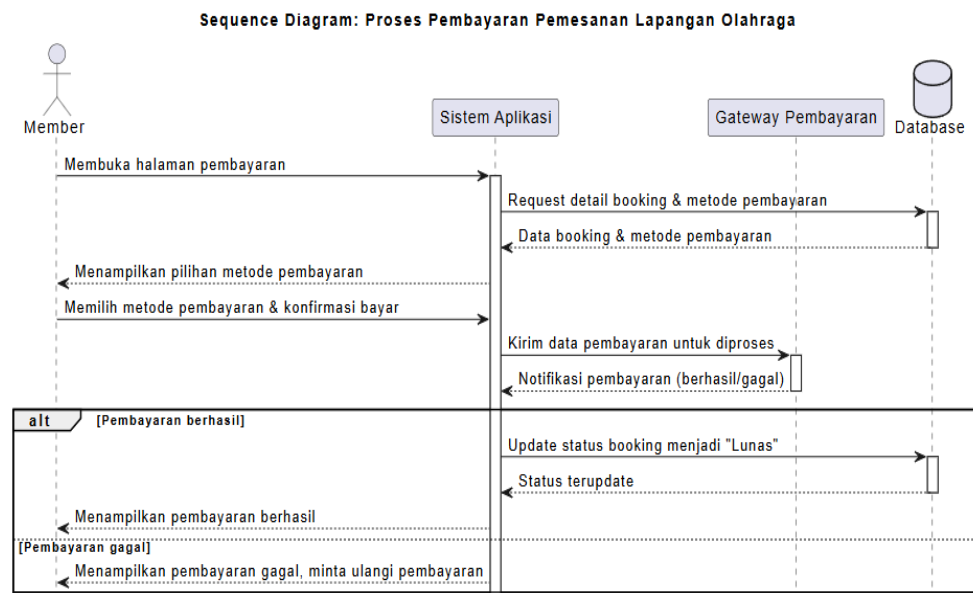


Figure 3. Sequence diagram of the field booking payment process

Deployment Diagram

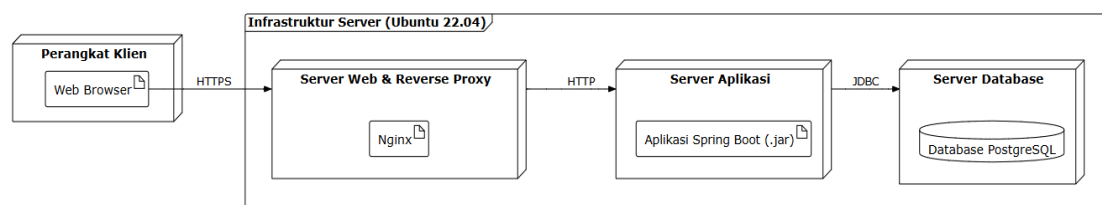


Figure 4. Deployment diagram of the sports field booking system

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) in the sports field booking system is used to model the data structure and relationships between key entities such as User, Booking, Field, Payment, Notification, and Maintenance. This ERD describes how a user can place orders and payments in the field, receive notifications, and how the field data and maintenance schedule

are managed in an integrated manner. With the existence of ERD, the database development process becomes more directed so that the integrity and consistency of data are maintained in the sports field booking system. (Pratama & Piliang, 2019)

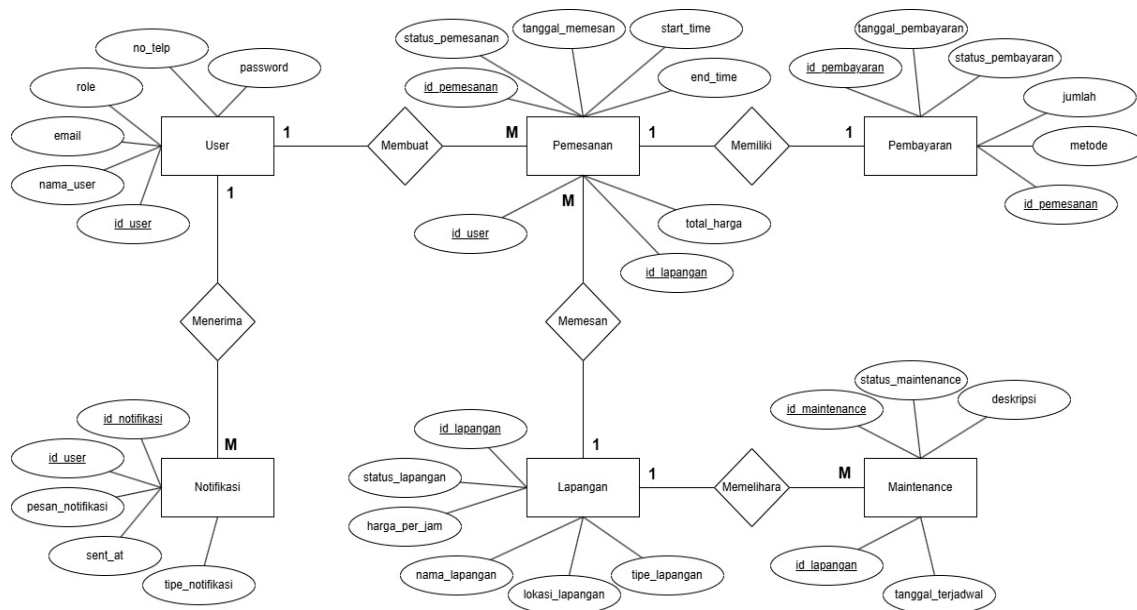


Figure 5. Entity relationship diagram (ERD) sports pitch booking system

Interface (UI) Design

The design of the interface (UI) in the web-based sports field booking information system is carried out by prioritizing ease of use (user friendly), clear navigation, and efficient access to information for users, both tenants and field managers. There are several main UI designs in this information system, namely: (1) The main page displays the venue search feature by name, city, and sport, as well as a list of venues along with rental prices in a structured manner for easy comparison and selection; (2) The login and registration page is designed to be simple and intuitive, allowing new and existing users to log in or create an account quickly, and provides a forgot password feature; (3) Field details and schedule selection page that provides complete information about venues, locations, hourly rental prices, and available schedule selection, so that users can directly make reservations as needed; (4) A schedule confirmation page that displays a summary of the schedule that the user has selected before proceeding to the next process; (5) Order and payment forms that display order details, fee details, and customer data to make the transaction process more transparent and easy; and (6) Report/admin dashboard that allows managers to monitor order, transaction, and report data in real-time. All visual elements are designed with the principle of minimalism, consistency, and responsiveness, making them convenient to access on various devices. (Stuttgart & Mardianto, 2019)

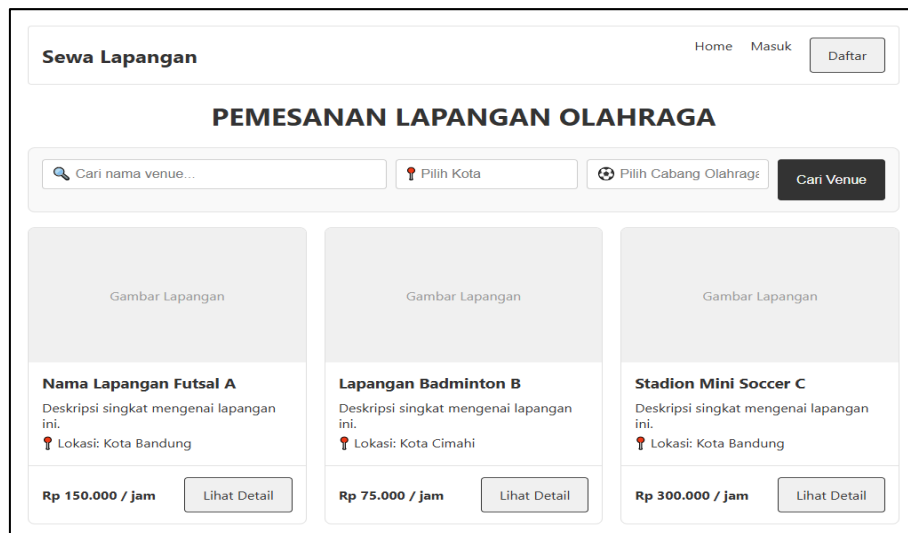


Figure 6. Sports pitch booking app interface view

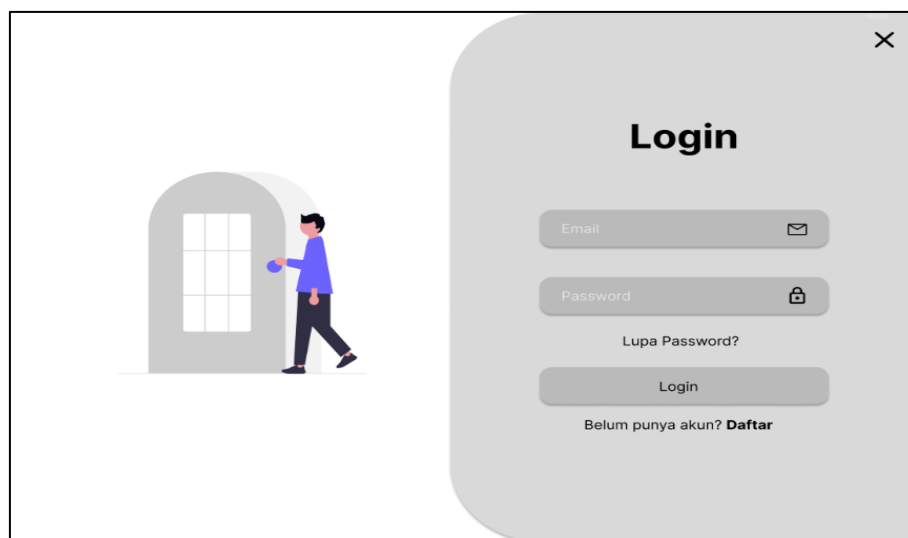


Figure 7. User login interface view

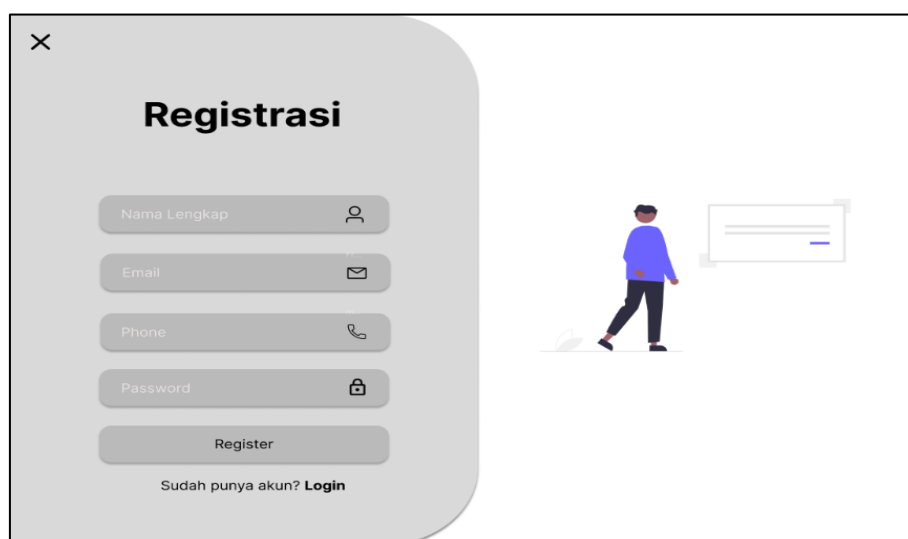


Figure 8. User registration interface view



Gambar Utama Lapangan

Nama Lapangan Futsal A

Lokasi: Jl. Olahraga No. 123, Kota Bandung

Deskripsi lengkap mengenai lapangan. Menjelaskan fasilitas yang tersedia seperti toilet, mushola, kafetaria, dan jenis lantai yang digunakan (sintetis/beton).

Fasilitas

- Toilet Bersih
- Mushola
- Parkir Luas
- Kafetaria

Pilih Jadwal

Pilih Tanggal

dd/mm/yyyy

Pagi (08:00 - 12:00)

08:00 09:00 10:00 11:00

Siang (12:00 - 18:00)

12:00 13:00 14:00 15:00

16:00 17:00

Malam (18:00 - 22:00)

18:00 19:00 20:00 21:00

Booking untuk 2 Jam

Figure 9. Field detail interface display and schedule selection

Pemesanan Lapangan Olahraga

Nama Pemesan

Nama Pengguna

Nomor Telepon

081234567890

Email

emailpengguna@mail.com

Ringkasan Pembayaran

Nama Lapangan Futsal A

Sabtu, 28 Oktober 2025

14:00 - 16:00 (2 Jam)

Harga Sewa (2 jam x Rp 150.000)	Rp 300.000
Biaya Layanan	Rp 5.000
Total Pembayaran Rp 305.000	

Lanjutkan Pembayaran

Figure 10. Interface view of the order form and payment summary

Konfirmasi Jadwal Pilihan

Anda telah memilih jadwal berikut untuk dipesan. Mohon periksa kembali sebelum melanjutkan.

Lapangan:	Nama Lapangan Futsal A
Tanggal:	Sabtu, 28 Oktober 2025
Waktu:	14:00 - 16:00
Durasi:	2 Jam

Batal Ya, Konfirmasi & Pesan

Figure 11. Pop-up interface view of selected schedules for field bookings

No	Kode Booking	Nama Lapangan	Tanggal	Customer	No Telp	Durasi	Total	Status	Action
1	21313	Lapangan Basket	01-06-2025	Unjani FC	0899999	180	150.000	Reserved	[Buttons]

Figure 12. Field booking data report interface view

Analysis of System Security Aspects

In addition to the functional design and interface, security is a fundamental aspect that is a priority in the development of this sports field booking information system. To ensure data integrity, confidentiality of user information, and service availability, a multi-layered security approach (Defense-in-Depth) is applied. This approach aligns with guidance from the Open Web Application Security Project (OWASP) that identifies a variety of critical risks to web applications. The security analysis of these systems includes data protection, common attack mitigation, access management, and infrastructure security. (Susilo, 2017) (Guaman et al., 2017)

User Data Protection

User data protection is a top priority in line with the obligations in the Personal Data Protection Law (UU PDP) No. 27 of 2022, which is implemented through two main mechanisms, namely data encryption in transit and data encryption when stored. Data encryption in transit is realized by using the SSL/TLS Certificate-supported HTTPS protocol, as shown in the Deployment Diagram (Figure 4), to ensure all communications between users and servers are encrypted so that sensitive data such as login credentials and booking details are protected from man-in-the-middle eavesdropping attacks. Meanwhile, encryption of data when stored is implemented through the use of powerful password hashing techniques such as Bcrypt, which automatically adds salt to each password before the hashing process, so that even if two users have the same password, the hash results in the database will be different and more resistant to rainbow table attacks in the event of a database breach (Sirohi et al., 2017) (Sriramya & Karthika, 2015) (Zhang et al., 2017).

Mitigating Common Attacks on Web Applications

Based on the risks identified by OWASP, several specific mitigations have been designed to improve system security, including protection against SQL Injection by utilizing modern frameworks such as Spring Boot and JPA that inherently use prepared statements or parameterized queries to separate SQL commands from user input data, thus preventing query manipulation by attackers. In addition, Cross-Site Scripting (XSS) prevention is done through proper encoding outputs so that the data displayed to the user has been sanitized so that

malicious scripts cannot be executed by the browser, as well as the implementation of HTTP Content Security Policy (CSP) headers as an additional layer to limit trusted script execution sources. To protect against Cross-Site Request Forgery (CSRF), frameworks such as Spring Security by default provide protection through the use of synchronization tokens, where any request that changes data must include a unique token that will be validated by the server, ensuring the request actually originates from a legitimate application interface. (Katole et al., 2018) (Yulianingsih, 2017) (Sudhodanan et al., 2017)

Access Management and Payment Security

Role-Based Access Control (RBAC) is implemented according to the identification of various user roles in the Use Case Diagram (Figure 1), such as Admin, Manager, and Member, each of which has different access rights to ensure each user can only access functions and data that correspond to their role (Principle of Least Privilege), thus preventing the occurrence of Broken Access Control vulnerabilities which are one of the highest risks in the OWASP Top 10 2021. In addition, the security of the Payment Gateway integration, as depicted in the Sequence Diagram (Figure 3), is maintained by never storing credit card data in the system to reduce the risk and scope of PCI DSS compliance, as well as implementing tokenization practices to have card-sensitive data replaced with unique, non-sensitive tokens. Payment notifications (webhooks) from the gateway are also validated using HMAC signatures to ensure the authenticity of the information and prevent fraudulent attempts through manipulation of payment data. (Cruz et al., 2018) (Woschek, 2015) (Santoso & others, 2016) (Jiang et al., 2016)

Infrastructure Security

The Deployment diagram (Figure 4) illustrates an architecture that supports infrastructure security, where Nginx acts as a reverse proxy that not only serves as a web server, but also serves as the first layer of defense by hiding the application server architecture, performing SSL terminations, as well as allowing the addition of security headers to strengthen protection. In addition, hardening servers using Ubuntu 22.04 is carried out by implementing standard security practices at the operating system level, such as configuring firewalls (e.g. UFW) to restrict access, regularly updating security, securing SSH access, and running application processes with minimum access rights to minimize potential exploits in the event of a compromise. (Purwaningrum et al., 2018)

By implementing this layered security approach, the designed sports pitch booking information system is not only functional and efficient, but also built on a strong and trusted security foundation.

CONCLUSION

This research successfully designed a comprehensive web-based sports field booking information system to address the inefficiencies and limitations of manual booking processes. By applying a literature review method and a structured system analysis and design approach, the study produced a system model that supports essential functionalities such as user registration, field searching, schedule selection, real-time booking with instant confirmation, and online payment integration. The proposed system design is supported by a web-based client-server architecture, a structured data model using an Entity Relationship Diagram (ERD), and clearly defined system interactions through use case and sequence diagrams. The

design also incorporates security considerations through measures such as data encryption, secure coding practices, role-based access control, and secure payment gateway integration. Overall, the system model is intended to improve administrative efficiency, reduce manual errors, and provide a more accessible and transparent booking experience for users. Future research is recommended to extend this work by implementing and testing the system in real-world environments, as well as exploring additional features such as mobile application integration, notification systems, user feedback modules, and data analytics to further enhance system functionality and user experience.

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